

Case Study: The education wage premium

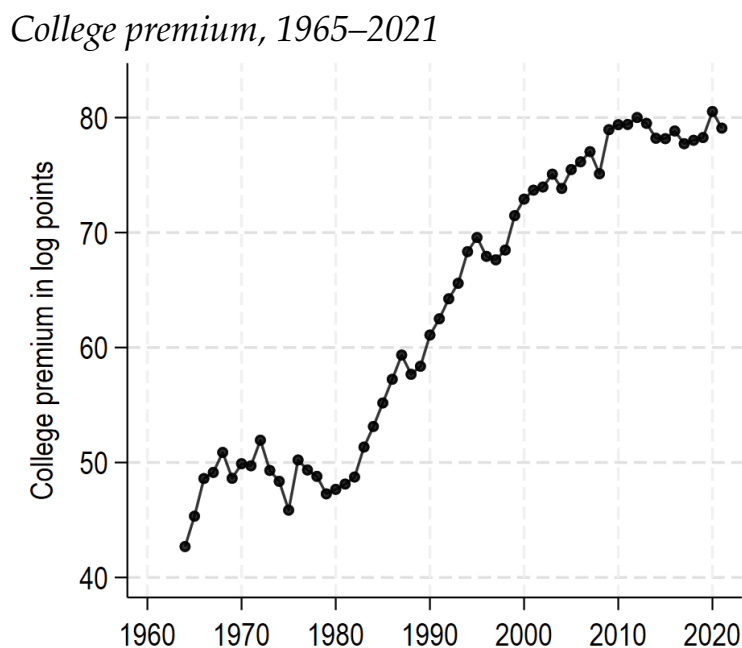
Computerization, task assignment, and the race between demand and supply

1 Context

Access to and takeup of education has increased consistently over time. During the last part of the 20th century and the beginning of the 21st century this resulted in an increasing share of university graduates. As more people went to university the share of non-college educated workers decreased. To put it (even more) formally, the graduate workforce became more *abundant* while the non-college educated workforce became more *scarce*. Yet the wages of university graduates increased relative to those of other workers during most of this period. That is, the *education wage premium* increased. The United States offers a useful historical puzzle. Its adjusted college wage premium rose substantially from around 1980 to 2010 and then became much flatter, as Exhibit 1 shows. (Canada has experienced a broadly similar trend.)

Exhibit 1. A wage premium that changes over time

United States: demographic-adjusted college wage premium, 1965–2021



Historical evidence. U.S. Current Population Survey wage data, adjusted for demographic differences. The vertical axis measures the college/non-college wage gap in log points. A higher value means a larger relative wage gap; it does not measure graduates' dollar wages or the net return to a degree. No conversion or numerical calculation is required.

The puzzle is about the balance between supply and demand: what changes in production could make employers want graduate labour even faster than its supply expands? Perhaps part of the puzzle is

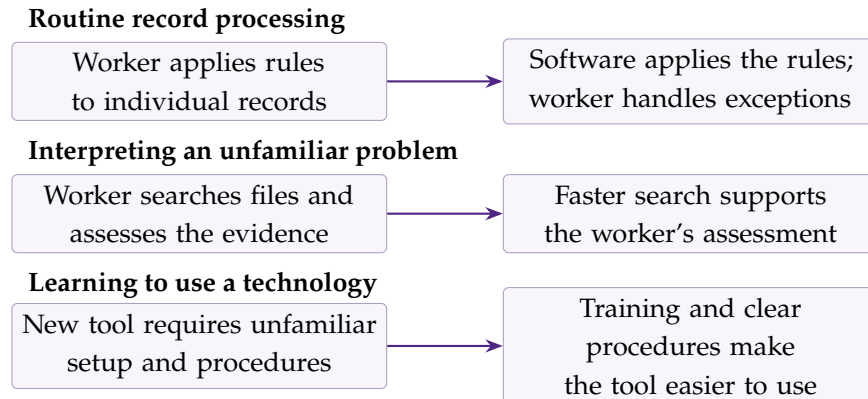
explained by what the education wage premium measures. As a relative measure of income, it can rise when both groups' wages increase, or when the lower-paid group's wage falls. It therefore tells us about inequality between groups, but does not by itself tell us whether either group is better off.

Perhaps the puzzle is explained by changes in technology. The earlier increase coincides with the first wave of computerization in the workplace. But saying that "technology favours graduates" leaves the central question unanswered: what do computers change about the work people perform?

Replacement and Assistance. Consider an office that prepares financial reports. Before computerization, employees spend time transferring records, applying accounting rules, checking totals, investigating unusual entries, and explaining results to clients. A program can take over some record processing and calculation. The same program can also make a worker more effective at investigating a discrepancy by making information easier to retrieve and compare. An employee's job can therefore change substantially even if its title stays the same.

These changes affect different activities within a job. For example routine and nonroutine tasks can be affected differently. A *routine* activity follows explicit, repeatable instructions; it need not be simple, unskilled, or manual. A *nonroutine* activity involves responding to circumstances that the program's rules do not cover. If graduates have a comparative advantage in the activities that expand, computerization can shift relative demand towards them.

Exhibit 2. Three ways a workplace can change



Learning and standardization. There are many other considerations that affect this discussion. For example, a new technology may initially need workers who can learn unfamiliar procedures, diagnose failures, and adapt it to a workplace. As instructions, training, and usable interfaces develop, more workers may become able to use it.

We see this in job advertisements data. The advertisements describe whether employers seeking users of a particular technology also ask for a college degree. Comparing these requirements as technologies age shows a pattern: newer technologies tend to have stronger college requirements, which diminish over time. However, even if each technology becomes easier to use, a continuing flow of new technologies can keep creating demand for workers with an initial learning advantage. If standardization catches up, that advantage may narrow while overall productivity continues to improve.

Keep separate two ways computerization can affect work: it may help workers perform tasks, or machines may take over tasks previously performed by workers. Questions 1–5 study how technology changes worker productivity and task assignment. A separate extension considers machines that perform tasks directly. The wage-premium graph alone cannot establish which mechanism mattered historically. Explaining the initial stability of the premium, its rise, and its plateau after 2010 requires more than assuming that a computer permanently raises one group's productivity once.

2 The CES model of production

Let H denote the supply of university graduates and L the supply of other workers, with wages w_H and w_L . The productivity of university graduates is A_H and that of other workers is A_L . Production is competitive, so each real factor reward equals its marginal product. The final-good price is P . Production technology is CES:

$$Y = \left[(A_H H)^{(\sigma-1)/\sigma} + (A_L L)^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad \sigma > 1. \quad (1)$$

With competitive production, each real wage equals the marginal product of its labour input:

$$\frac{w_H}{P} = A_H^{(\sigma-1)/\sigma} \left(\frac{Y}{H} \right)^{1/\sigma}, \quad \frac{w_L}{P} = A_L^{(\sigma-1)/\sigma} \left(\frac{Y}{L} \right)^{1/\sigma}.$$

The income shares are

$$\alpha_H = \frac{w_H H}{PY}, \quad \alpha_L = \frac{w_L L}{PY}, \quad \alpha_H + \alpha_L = 1.$$

The education wage premium is given by the relative wage

$$\frac{w_H}{w_L} = \left(\frac{A_H}{A_L} \right)^{(\sigma-1)/\sigma} \left(\frac{H}{L} \right)^{-1/\sigma} \rightarrow \Delta \log \frac{w_H}{w_L} = \frac{\sigma-1}{\sigma} \Delta \log \frac{A_H}{A_L} - \frac{1}{\sigma} \Delta \log \frac{H}{L}.$$

The effects of one group's productivity on the other group's real wage are

$$\frac{\partial \log(w_L/P)}{\partial \log A_H} = \frac{\alpha_H}{\sigma} > 0, \quad \frac{\partial \log(w_H/P)}{\partial \log A_L} = \frac{\alpha_L}{\sigma} > 0.$$

Question 1. Can graduate supply alone explain the rising premium?

(20 marks)

Between 1980 and 2010, the relative supply H/L rose while Exhibit 1 shows a rising education wage premium. University administrators claim that the greater relative supply of graduates, by itself, caused their relative wage to rise (everyone should go to college! the more the merrier!). Evaluate the claim using the CES relative-wage equation. Draw a relative-demand and relative-supply diagram, showing the initial equilibrium and the equilibrium predicted by a supply increase alone. Explain how this prediction compares with Exhibit 1.

Question 2. What must have happened to relative demand?

(25 marks)

(a) Show in your Question 1 diagram the relative-demand change needed to produce a higher w_H/w_L despite the increase in H/L . Give an example of a change in the way production takes place in the workplace that is consistent with your explanation. [15 marks]

(b) Use the CES relative-wage equation to state the lower bound on the change in A_H/A_L needed for the premium to rise. Express your answer as an inequality in log changes, and explain why it is a statement about the change rather than the level of relative productivity. [10 marks]

Question 3. Which productivity changes fit the facts?

(25 marks)

(a) Suppose computerization raises A_H alone. People are worried that this will drive down the real wage of workers without a university degree. Show what happens to w_H/w_L and w_L/P and discuss the worries about workers' wages. [10 marks]

(b) Suppose standardization raises A_L alone. What happens to the premium and w_H/P ? How does the predicted premium change relate to the case's description of older technologies? [8 marks]

(c) Suppose both productivity terms rise in the same proportion. What happens to the premium and both real wages? Evaluate the claim that the introduction of personal computers around 1980 was a one-time event that led to the continuous increase in the education wage premium? [7 marks]

3 How task assignment generates the CES

Suppose there is a continuum of tasks indexed by $i \in [0, 1]$. Each task produces a service $y(i)$. The final good combines these services with equal-weight Cobb–Douglas aggregation:

$$\log Y = \int_0^1 \log y(i) di \quad \text{where} \quad y(i) = a_L(i)L + a_H(i)H.$$

Either labour group can perform any task, with productivity

$$a_L(i) = A_L(1-i)^{1/\kappa}, \quad a_H(i) = A_H i^{1/\kappa}, \quad \kappa > 1.$$

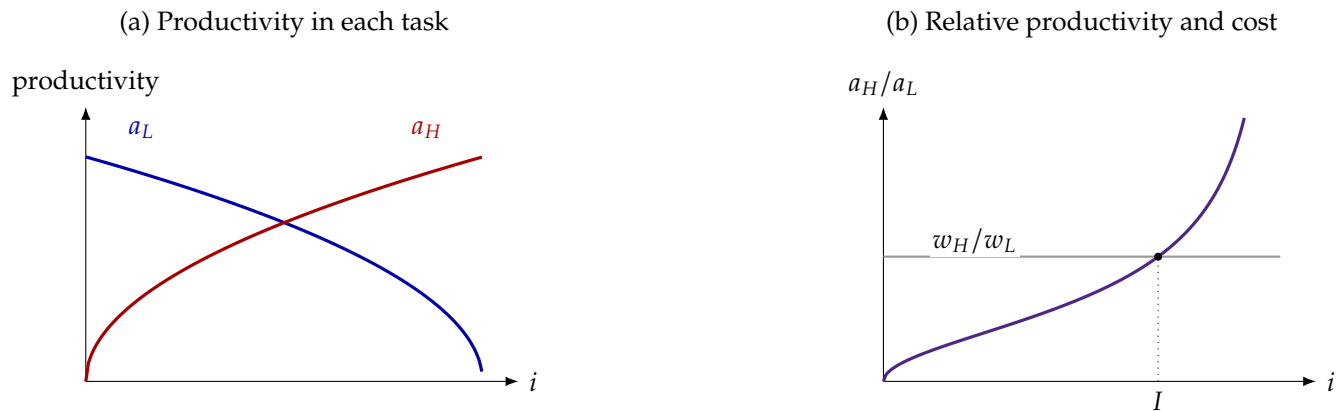
For the task to be assigned to L or H , compare unit costs $w_L/a_L(i)$ and $w_H/a_H(i)$. Their relative productivity is

$$\frac{a_H(i)}{a_L(i)} = \frac{A_H}{A_L} \left(\frac{i}{1-i} \right)^{1/\kappa}.$$

It rises with i : L has comparative advantage at low-index tasks and H at high-index tasks. The cutoff I makes the two unit costs equal; L performs tasks below I and H performs those above it. The cutoff is the solution to:

$$\frac{I}{1-I} = \left(\frac{w_H/A_H}{w_L/A_L} \right)^\kappa.$$

The figure shows the productivity schedules in panel (a) and the relative productivity line in panel (b).

Figure 1: Task productivities and relative productivity with high κ 

With both labour supplies fully employed, equal task expenditure per unit interval gives

$$w_L L = IPY, \quad w_H H = (1 - I)PY.$$

Thus $\alpha_L = I$ and $\alpha_H = 1 - I$ in this task model.

Solving task assignment and labour-market clearing gives back the CES production function with $\sigma = 1 + \kappa$:

$$Y = e^{-1/\kappa} \left[(A_H H)^{\kappa/(1+\kappa)} + (A_L L)^{\kappa/(1+\kappa)} \right]^{(1+\kappa)/\kappa}.$$

Question 4. Do graduates have an advantage in every task? (15 marks)

Assume $A_H > A_L$ and $w_H > w_L$. One manager claims that the firm should hire only university graduates and assign them every task because they know more and are better prepared, as $A_H > A_L$ shows. A second manager says the first manager is wrong: graduates are less productive in low-index tasks and should be assigned only tasks in which their productivity is actually higher $a_H(i) > a_L(i)$. State which tasks each group should perform and support your conclusions formally, use this results to evaluate the claims of the two managers.

Question 5. Interpret the wage-premium history with the task model (15 marks)

The case describes computer assistance, and Exhibit 1 shows a rising premium followed by a plateau after 2010. Suppose computer assistance raises A_H while H , L , A_L , and κ stay fixed. As wages adjust, what happens the premium and the task cutoff? Does the wage response undo graduates' initial task expansion? Use the model relationships to support your answer. Can Exhibit 1 establish that this mechanism caused the rise or explain why the premium later flattened?

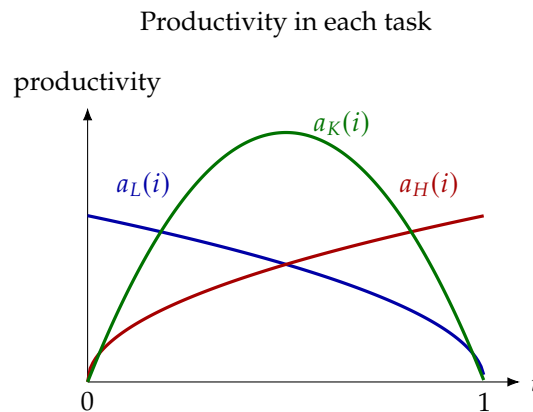
4 Optional extension: Machines perform middle tasks

Now suppose a newer technology makes machines able to perform tasks on their own with productivity

$$a_K(i) = A_K i(1 - i), \quad A_K > 0.$$

Thus $a_K(0) = a_K(1) = 0$, while machine productivity is positive in the interior and highest in the middle as in Figure 2. Firms now assign tasks between workers and machines choosing the lowest unit cost at each task. Machine services are available at rental price r .

Figure 2: A machine with comparative strength in middle tasks



For the extension question, compare an initial allocation without machine production to a later allocation in which machines are the cheapest producer on one interior interval (J_L, J_H) with $J_L < I < J_H$. In the later allocation, L performs $i < J_L$ and H performs $i > J_H$. The supplies L, H remain fixed and fully employed; wages adjust. The initial worker cutoff is I from the earlier task model. The cutoffs satisfy the same lowest-unit-cost rule and equal expenditure per unit task interval to all available producers.

Extension question. Can middle-task automation explain the education premium?

- (a) At given factor prices, write the unit-cost condition for a machine to perform task i rather than either worker group. Explain why the hump in Figure 2 does not by itself guarantee automation, why tasks near the endpoints stay with workers, and how a fall in r/A_K affects the automated interval.
- (b) Write each group's wage bill before and after automation (use task expenditures define above). Derive the necessary and sufficient condition for the education wage premium to rise at fixed H, L . Can replacement of middle tasks account for the rise in Exhibit 1 without that condition?
- (c) Which workers benefit from automation via higher real wages? To answer, express each group's real wage before and after automation using output and its task-expenditure share and state which real wage rises and which falls. Assume that the change in output satisfies

$$\frac{1 - I}{1 - J_H} < \frac{Y_1}{Y_0} < \frac{I}{J_L}.$$

References

This case study is based on models, results, and exhibits from the following papers. The questions and classroom illustrations were prepared for this course.

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- Autor, David H., Frank Levy, and Richard J. Murnane (2003). *The Skill Content of Recent Technological Change: An Empirical Exploration*. *Quarterly Journal of Economics* 118(4), 1279–1333.
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